In [4]:
```
# import pandas library
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
from sklearn.linear_model import LogisticRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
# Reading csv file
df = pd.read_csv("Heart.csv")
df.head()
```

Out[3]:

	age	sexe	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targ
0	64	1	0	128	263	0	1	105	1	0.2	1	1	3	
1	46	0	2	142	177	0	0	160	1	1.4	0	0	2	
2	65	0	2	140	417	1	0	157	0	0.8	2	1	2	
3	66	1	0	160	228	0	0	138	0	2.3	2	0	1	
4	57	1	0	165	289	1	0	124	0	1.0	1	3	3	

In [5]:
```
df = df.drop_duplicates()
```

In [6]:
```
# Count ,min,max ,etc of each column
df.describe()
```

Out[6]:

	age	sexe	cp	trestbps	chol	fbs	restecg	
count	212.000000	212.000000	212.000000	212.000000	212.000000	212.000000	212.000000	2
mean	54.731132	0.679245	1.000000	132.033019	244.976415	0.160377	0.514151	1
std	9.159377	0.467871	1.066509	17.370887	53.194665	0.367824	0.519558	
min	29.000000	0.000000	0.000000	94.000000	131.000000	0.000000	0.000000	
25%	48.000000	0.000000	0.000000	120.000000	209.000000	0.000000	0.000000	1
50%	56.000000	1.000000	1.000000	130.000000	237.500000	0.000000	1.000000	1
75%	62.000000	1.000000	2.000000	140.000000	271.750000	0.000000	1.000000	1
max	76.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	2

In [7]:
```
# Information about each column data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 212 entries, 0 to 211
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       212 non-null    int64
 1   sexe      212 non-null    int64
 2   cp        212 non-null    int64
 3   trestbps  212 non-null    int64
 4   chol      212 non-null    int64
 5   fbs       212 non-null    int64
 6   restecg   212 non-null    int64
 7   thalach   212 non-null    int64
 8   exang     212 non-null    int64
 9   oldpeak   212 non-null    float64
 10  slope     212 non-null    int64
 11  ca        212 non-null    int64
 12  thal      212 non-null    int64
 13  target    212 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 24.8 KB
```

In [8]:
```
#Finding null values in each column
df.isna().sum()# Information about each column data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 212 entries, 0 to 211
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       212 non-null    int64
 1   sexe      212 non-null    int64
 2   cp        212 non-null    int64
 3   trestbps  212 non-null    int64
 4   chol      212 non-null    int64
 5   fbs       212 non-null    int64
 6   restecg   212 non-null    int64
 7   thalach   212 non-null    int64
 8   exang     212 non-null    int64
 9   oldpeak   212 non-null    float64
 10  slope     212 non-null    int64
 11  ca        212 non-null    int64
 12  thal      212 non-null    int64
 13  target    212 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 24.8 KB
```

In [9]:
```
df.head()
```

Out[9]:

	age	sexe	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targ
0	64	1	0	128	263	0	1	105	1	0.2	1	1	3	
1	46	0	2	142	177	0	0	160	1	1.4	0	0	2	
2	65	0	2	140	417	1	0	157	0	0.8	2	1	2	
3	66	1	0	160	228	0	0	138	0	2.3	2	0	1	
4	57	1	0	165	289	1	0	124	0	1.0	1	3	3	

In [10]:
```
df.fbs.unique()
```

Out[10]:
```
array([0, 1])
```

In [12]:
```
subSet1 = df[['age','cp','chol','thalach']]
```

In [15]:
```
subSet2 = df[['exang','slope','target']]
```

In [16]:
```
merged_df = subSet1.merge(right=subSet2,how='cross')
merged_df.head()
```

Out[16]:

	age	cp	chol	thalach	exang	slope	target
0	64	0	263	105	1	1	1
1	64	0	263	105	1	0	1
2	64	0	263	105	0	2	1
3	64	0	263	105	0	2	1
4	64	0	263	105	0	1	0

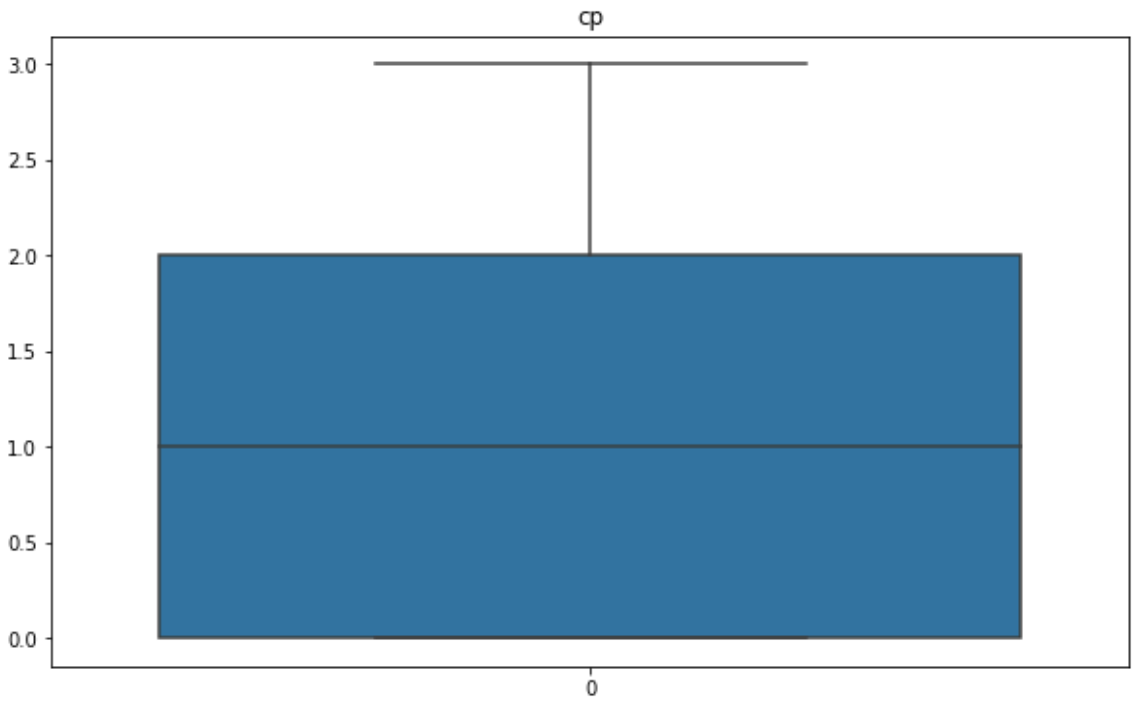
In [17]:
```
df.columns
```

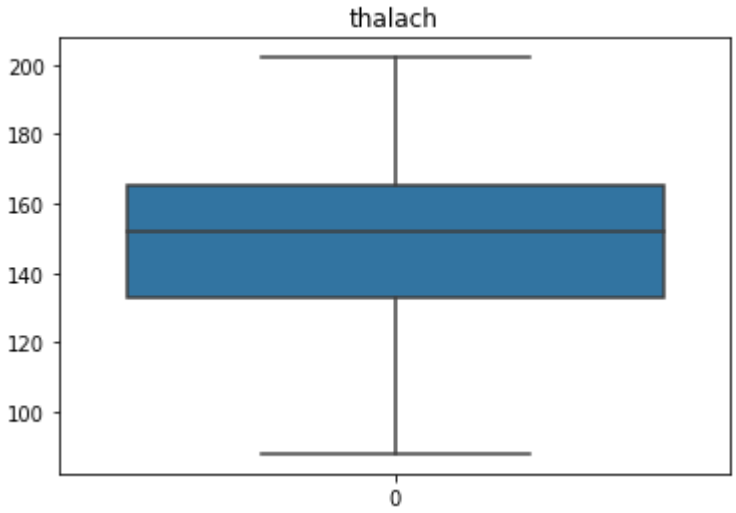
Out[17]:
```
Index(['age', 'sexe', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalac
h',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

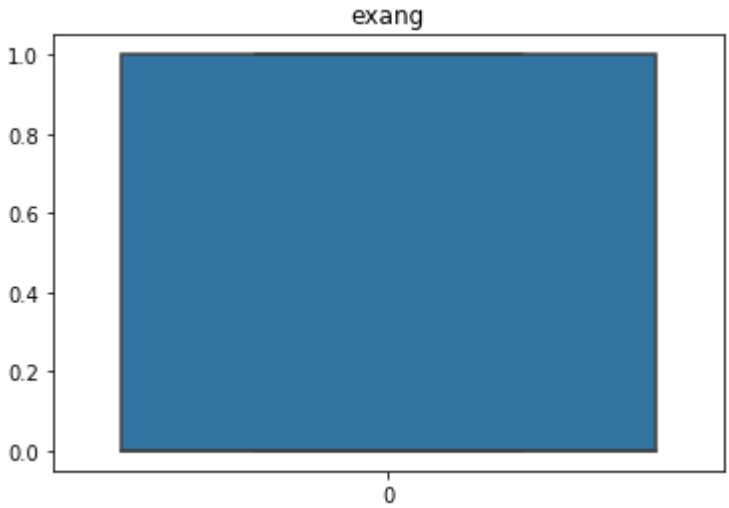
In [18]:
```
def remove_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    threshold = 1.5 * IQR
    outlier_mask = (column < Q1 - threshold) | (column > Q3 + threshold)
    return column[~outlier_mask]
```

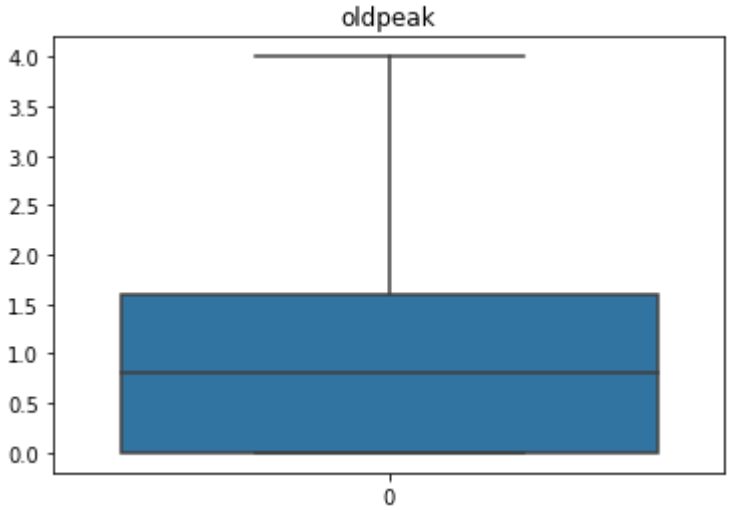
In [20]:
```
# Remove outliers for each column using a loop
col_name = ['cp','thalach','exang','oldpeak','slope','ca']
for col in col_name:
    df[col] = remove_outliers(df[col])
```

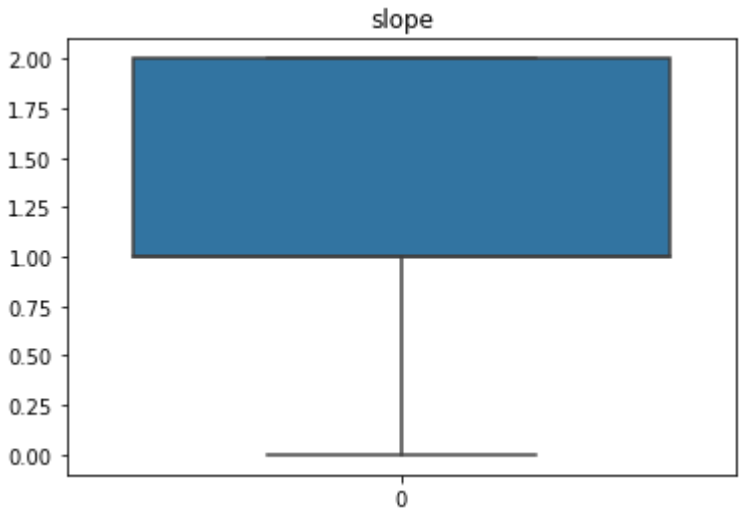
In [21]:
```
plt.figure(figsize=(10, 6))  # Adjust the figure size if needed

for col in col_name:
    sns.boxplot(data=df[col])
    plt.title(col)
    plt.show()
```

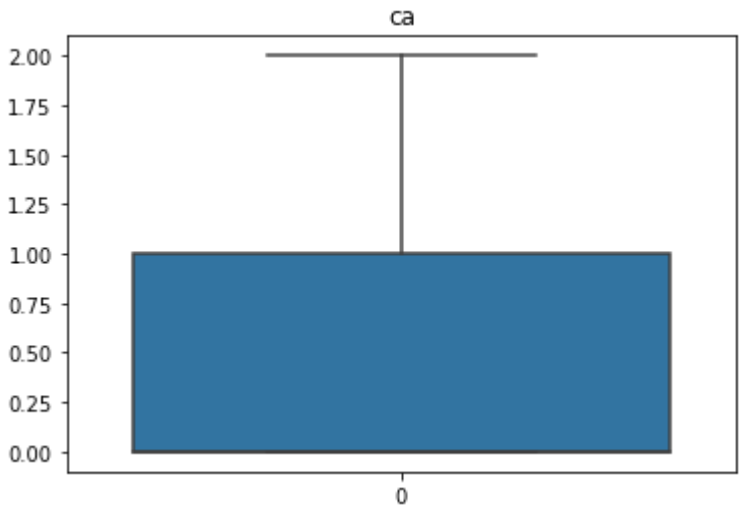

cp
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0


thalach
200
180
160
140
120
100
0


exang
1.0
0.8
0.6
0.4
0.2
0.0
0







In [22]:
```
df = df.dropna()
```

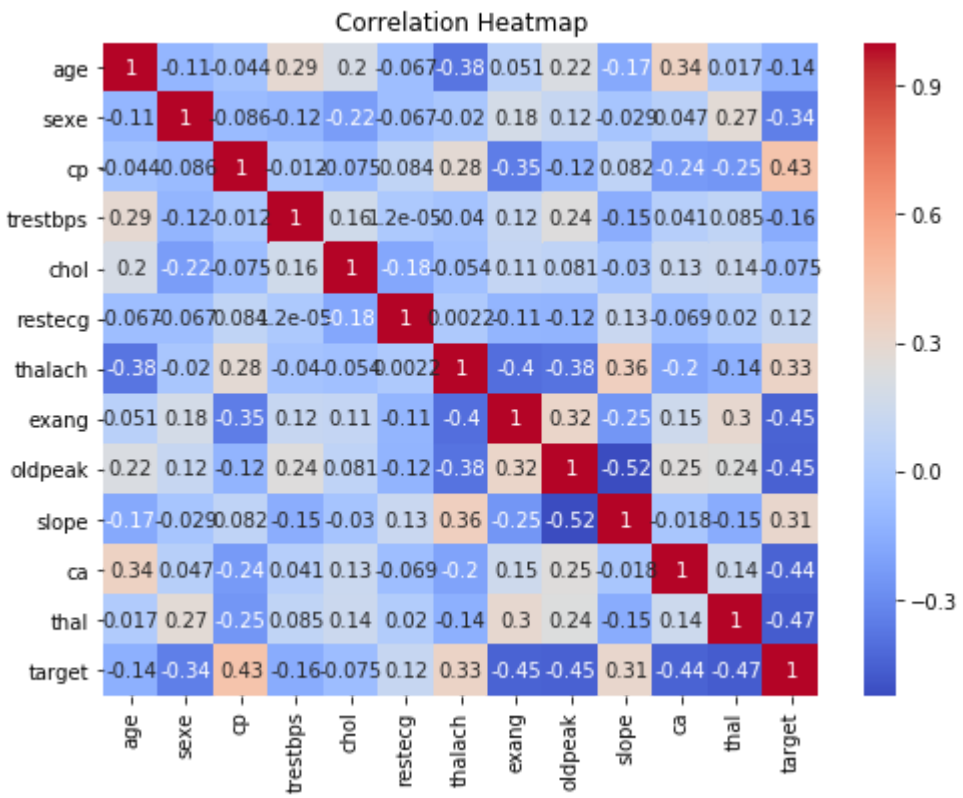
In [23]:
```
df.isna().sum()
```

Out[23]:
```
age         0
sexe        0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

In [24]:
```
df = df.drop('fbs',axis=1)
```

In [26]:
```
# Compute correlations between features and target
correlations = df.corr()['target'].drop('target')

# Print correlations
print("Correlation with the Target:")
print(correlations)
print()

# Plot correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
Correlation with the Target:
age        -0.141383
sexe       -0.342506
cp          0.432323
trestbps   -0.155796
chol       -0.074646
restecg     0.116095
thalach     0.328455
exang      -0.445435
oldpeak    -0.448892
slope       0.306024
ca         -0.441641
thal       -0.468544
Name: target, dtype: float64
```



In [27]:
```
# df.isna().sum()
```

In [30]:
```
# splitting data using train test split
x = df[['cp','thalach','exang','oldpeak','slope','ca']]
y = df.target
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_sta
te=0)

x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

Out[30]:
```
((153, 6), (39, 6), (153,), (39,))
```

In [31]:
```
from sklearn.preprocessing import StandardScaler
```

In [32]:
```
scaler = StandardScaler()
```

In [33]:
```
x_train_scaled = scaler.fit_transform(x_train)
x_test_scaled = scaler.transform(x_test)
```

```
/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/preprocessing/
data.py:645: DataConversionWarning: Data with input dtype int64, float64 w
ere all converted to float64 by StandardScaler.
  return self.partial_fit(X, y)
/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/base.py:464: D
ataConversionWarning: Data with input dtype int64, float64 were all conver
ted to float64 by StandardScaler.
  return self.fit(X, **fit_params).transform(X)
/home/rmdstic/anaconda3/lib/python3.7/site-packages/ipykernel_launcher.py:
2: DataConversionWarning: Data with input dtype int64, float64 were all co
nverted to float64 by StandardScaler.
```

In [34]:
```
y_train= np.array(y_train).reshape(-1, 1)
y_test= np.array(y_test).reshape(-1, 1)
```

In [35]:
```
y_train.shape
```

Out[35]:
```
(153, 1)
```

In [36]:
```
model = LogisticRegression()
model.fit(x_train_scaled, y_train)

# Make predictions on the test set
y_pred = model.predict(x_test_scaled)

# Evaluate the model's accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/linear_model/l
ogistic.py:433: FutureWarning: Default solver will be changed to 'lbfgs' i
n 0.22. Specify a solver to silence this warning.
  FutureWarning)
/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/utils/validati
on.py:761: DataConversionWarning: A column-vector y was passed when a 1d a
rray was expected. Please change the shape of y to (n_samples, ), for exam
ple using ravel().
  y = column_or_1d(y, warn=True)

Accuracy: 0.8974358974358975

/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/linear_model/b
ase.py:283: DeprecationWarning: `np.int` is a deprecated alias for the bui
ltin `int`. To silence this warning, use `int` by itself. Doing this will
not modify any behavior and is safe. When replacing `np.int`, you may wish
to use e.g. `np.int64` or `np.int32` to specify the precision. If you wish
to review your current use, check the release note link for additional inf
ormation.
Deprecated in NumPy 1.20; for more details and guidance: https://numpy.or
g/devdocs/release/1.20.0-notes.html#deprecations
  indices = (scores > 0).astype(np.int)
```

In [37]:
```
#Classification model using Decision Tree
from sklearn.tree import DecisionTreeClassifier
tc=DecisionTreeClassifier(criterion='entropy')
tc.fit(x_train_scaled,y_train)
y_pred=tc.predict(x_test_scaled)

print("Training Accuracy Score :",accuracy_score(y_pred,y_test))
print("Training Confusion Matrix  :",confusion_matrix(y_pred,y_test))
```

```
Training Accuracy Score : 0.717948717948718
Training Confusion Matrix  : [[12  7]
 [ 4 16]]

/home/rmdstic/anaconda3/lib/python3.7/site-packages/sklearn/tree/tree.py:1
49: DeprecationWarning: `np.int` is a deprecated alias for the builtin `in
t`. To silence this warning, use `int` by itself. Doing this will not modi
fy any behavior and is safe. When replacing `np.int`, you may wish to use
e.g. `np.int64` or `np.int32` to specify the precision. If you wish to rev
iew your current use, check the release note link for additional informati
on.
Deprecated in NumPy 1.20; for more details and guidance: https://numpy.or
g/devdocs/release/1.20.0-notes.html#deprecations
  y_encoded = np.zeros(y.shape, dtype=np.int)
```

In [ ]: